SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 6

Process Design and Analysis

Process design and analysis (Ch. 7 manufacturing, Ch. 9 services) plus BPMN 2.0 and the service blueprint, with the KAI case — One Second at the Gambir Gate (and references to Albright and Groom). Covered: processes and why type matters, the customer order decoupling point, five layouts, the product–process matrix, break-even equipment choice (640 and 2,000 units), flow-design tools, customer contact and the service-system design matrix, customer-introduced variability and four accommodation strategies, service poka-yokes (Three Ts), Ritz-Carlton scripting, three service designs, BPMN 2.0, the service blueprint and the KAI discussion.

Material

1 Learning objectives and roadmap

1. Understand manufacturing processes and the customer order decoupling point.
2. Read the product–process and service-system design matrices.
3. Analyse simple processes: break-even, flow design, bottlenecks.
4. Model with BPMN 2.0 and the service blueprint.
5. Apply to live cases.

2 What a process is, and why the type matters

- Manufacturing converts inputs into a physical output; in services the customer is inside the system.
- Process selection is driven by volume and standardisation.
- Every process has a capacity, bottleneck and flow time, fixed by design.

- What matters is where the customer sits relative to the work.

Manufacturing
Inventory absorbs mismatch.

vs

Service
Capacity is perishable.

3 The customer order decoupling point

The decoupling point is where inventory lets one part of the process run independently of another — how far the customer order reaches in.

Environment	How	Example
Make-to-stock	From finished goods	TVs, clothing, packaged food
Assemble-to-order	Modules combined	Dell: 17 components → 384 configurations
Make-to-order	From raw materials	Boeing aircraft
Engineer-to-order	Designed with the customer	Lockheed Martin satellites

> **Strategic**
> Closer to the customer = faster but costlier inventory; moving upstream trades speed for cost.

4 Five layouts and the product–process matrix

Layout	Flow	Example
Project	Product fixed, resources come to it	Construction, film lots
Workcenter	Similar machines grouped	Low volume, high variety

Manufacturing cell	Dissimilar machines for one family	Metal fabrication, chips
Assembly line	Progressive steps, paced	High volume
Continuous	Uninterrupted flow	Refinery

Product–process matrix: volume × standardisation, processes on the diagonal; off-diagonal is expensive (mass customisation top right); cells span the widest range.

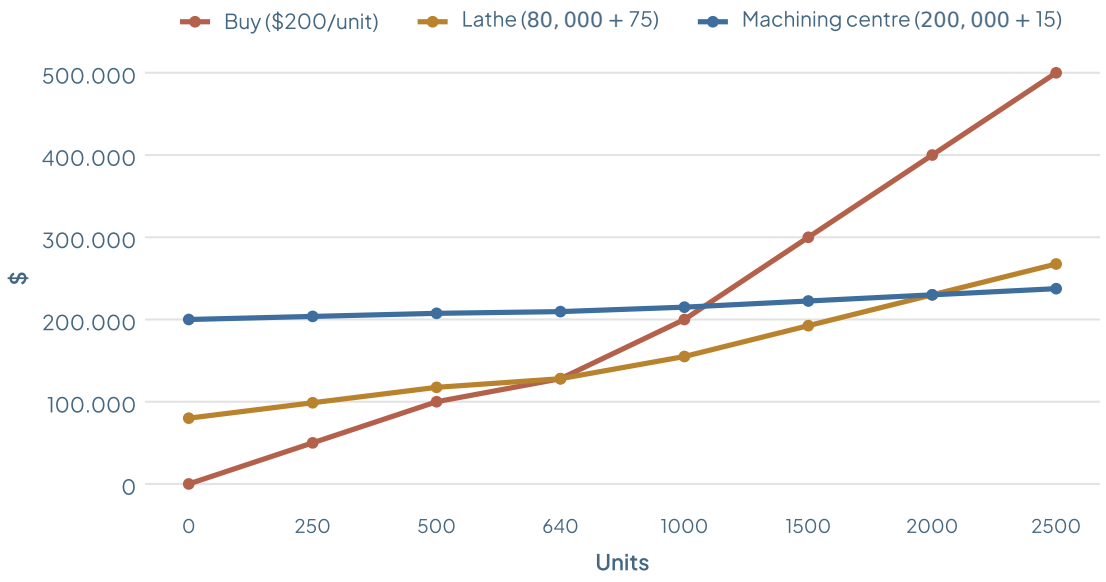
5 Break-even: choosing the equipment (Example 7.1)

Option	Total cost
Buy	`$200 × D`
Semiautomatic lathe	`$80,000 + $75 × D`
Machining centre	`$200,000 + $15 × D`

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B: 80,000 ÷ 125 = 640 units · A: 120,000 ÷ 60 = 2,000 units
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Below 640 buy; 640–2,000 lathe; above 2,000 machining centre.

Total cost of three options (Example 7.1)



6 Manufacturing process flow design

Tool	Content
Assembly drawing	Exploded view of components
Assembly chart	How parts go together, order, material flow
Operation & route sheet	Operations, equipment, tooling per part
Flow process chart	Documents the flow; steady-state diagnosis

Why chart

You cannot improve a process you have not drawn.

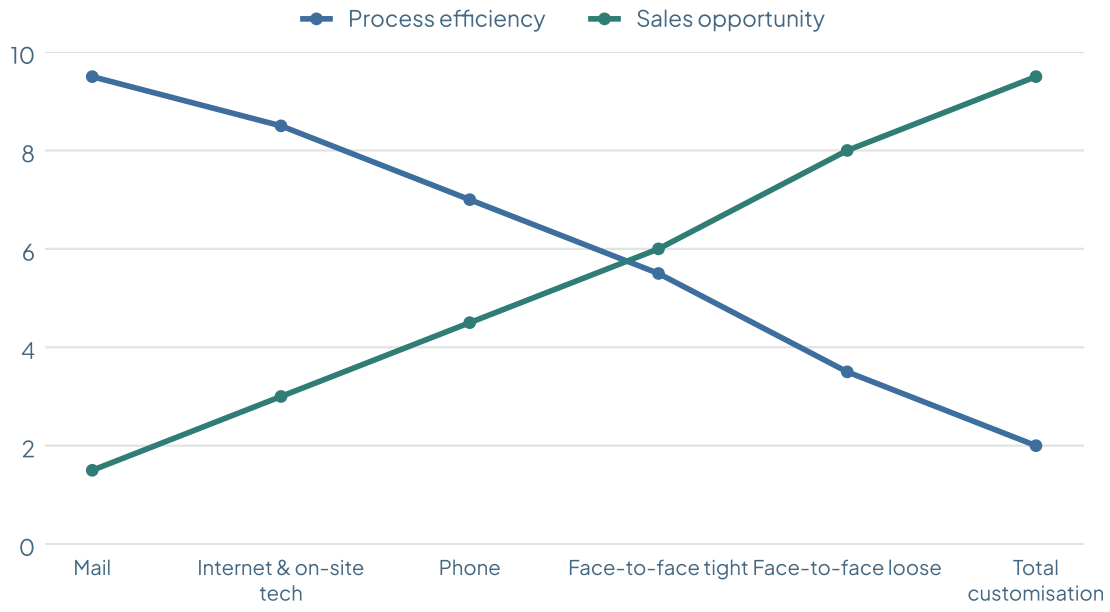
Plug example: components (circles) merge into subassemblies SA-1–SA-4, then A-1 and A-2, then the finished plug.

7 Customer contact and the service-system design matrix

Extent of contact = % of time the customer must be in the system. High contact: customer affects timing, nature and perceived quality; diagnostic skills; capacity and client mix; high variability. Low contact: back-office work on surrogates; designed like a factory; efficient.

Dimension	Low	Mixed	High
Example	Cheques, mail order	Call centre, teller	Clinic, consulting
Skills	Clerical	Verbal, procedural	Diagnostic, trade
Efficiency	High	Moderate	Low
Sales opportunity	Low	Moderate	High

Contact vs efficiency & sales opportunity (conceptual)



Service-system design matrix: buffered core (mail), permeable (internet/on-site tech, phone, face-to-face tight specs), reactive (face-to-face loose

specs, total customisation). Efficiency falls and sales opportunity rises with contact.

8 Customer-introduced variability and four accommodation strategies

Variability	Classic accommodation	Low-cost accommodation	Classic reduction	Uncompromise reduction
Arrival	Extra staff at peaks	Part-time labour, off-peak pricing	Reservations	Segment by arrival pattern
Request	Broader skills, wider menu	Self-service configuration	Restrict the offer	Persuade towa standard
Capability	Extra help	Peer support	Screen before entry	Design so capability doesn't matter
Effort	Do the work for them	Incentives	Penalties	Make the desire action easiest
Subjective preference	More employees, attention	Self-select	Set expectations	Read and adap style

Point

Uncompromised reduction breaks the cost–quality framing (Redbox charges per day).

9 Service poka-yokes, scripted processes and three service designs

Poka-yokes block mistakes from becoming service defects, often fail-safing the customer too.

Three Ts	Question	Example
Task	Was the car fixed right?	Indented surgical trays; kitting

Treatment	Was the manager courteous?	Mirrors on phones; scripted greetings
Tangibles	Was the area clean?	Height bars; pictures of a clean room

Ritz-Carlton's three steps: warm welcome; anticipation and compliance (identify, fulfil, verify; break away for complaints); fond farewell — with preferences logged to guest history.

Design	Example	Matrix position
Production-line	McDonald's	Face-to-face tight specs
Self-service	ATMs, gas stations	Internet & on-site tech
Personal-attention	Nordstrom, Ritz-Carlton	Total customisation

10 BPMN 2.0: core vocabulary and when to use it

Family	Content
Activities	Task, sub-process, transaction, call activity
Events	Start, intermediate, end; message, timer, error, escalation, signal, conditional
Gateways	Exclusive, parallel, inclusive, event-based, complex
Pools & lanes	Participants and roles; message flow crosses pools
Data	Objects, stores

> **Gateways**
> Exclusive = one branch; parallel = all branches; event-based = first event wins. Common error: parallel where exclusive belongs.

BPMN adds ownership, exception paths, executability and a shared grammar; route sheets still win for purely physical flows inside one plant.

≡ Albright escalation

The "attach context to transfer" task is a mandatory step — a poka-yoke in notation.

11 The service blueprint: five layers, three lines

Layer	Content
Evidence	Artefacts the customer encounters
Customer journey	Steps and emotions
Frontstage	Visible employee actions and tech
Backstage	Invisible actions the encounter depends on
Support processes	Systems, suppliers, internal functions

Lines: interaction (moments of truth), visibility, internal interaction. Use BPMN for handoffs and exceptions; a blueprint for experience failures; serious redesigns use both.

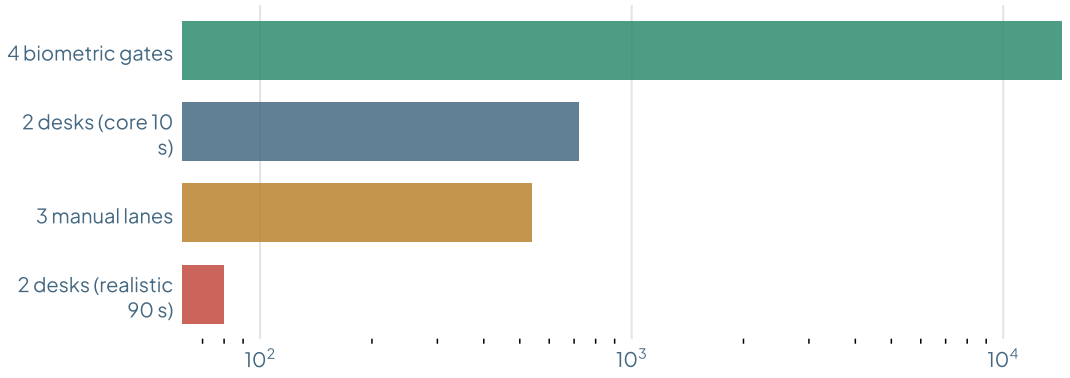
12 The KAI case — One Second at the Gambir Gate

KAI (503.5m customers in 2025, +8.52%) must decide whether to expand face-recognition gates (1-second verification; 5.55m users in H1 2026; 22 → 60 stations). The Bandung Station Master: the queue is at registration, not the gate. KAI lacks registration time, ID read-failure and fallback rates.

! Arithmetic

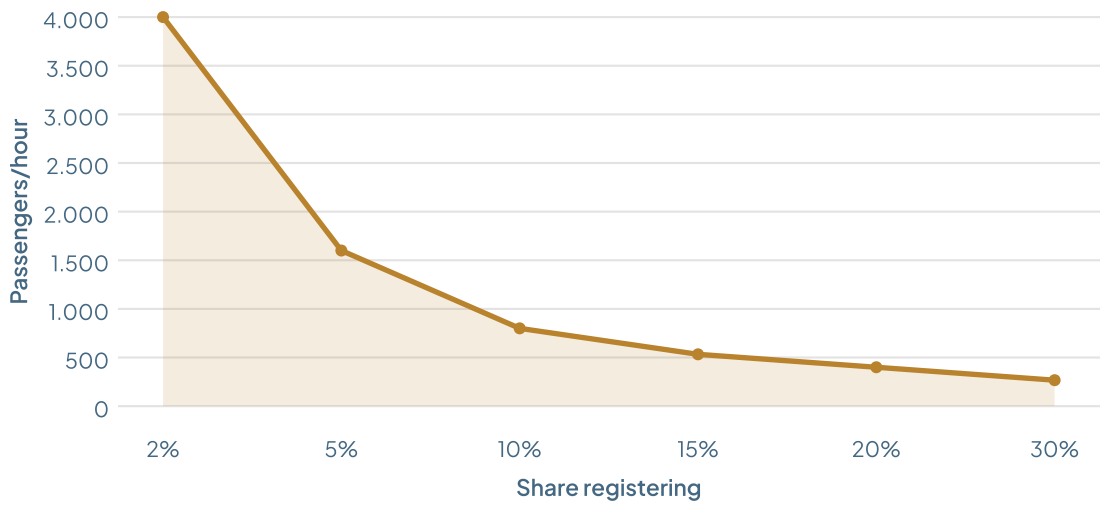
Gate ~14,400/hour vs registration desk ~80/hour.

Hourly capacity per resource (log scale)




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Ceiling = registration capacity ÷ share registering at the station
= 80 ÷ p
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Passenger ceiling vs share registering at the station



Discussion: the failure is a support process pushed into the journey; measure registration time, read failures and fallback share at peak before expanding; the manual path needs a capacity standard to be real; baseline before buying.

13 Key takeaways from the KAI case

1. Measure the whole path, not the fastest point.
2. A support layer pushed to the surface.

3. Failure is felt at the line of interaction.
4. Capability variability spikes at peak.
5. Success figures are not evidence — baseline first.
6. A public mandate changes the objective; availability needs a capacity standard.

Key takeaways

1. Customer inside the system; volume and standardisation drive process choice.
2. Decoupling point: MTS, ATO, MTO, ETO.
3. Five layouts.
4. Product–process matrix diagonal.
5. Break-even at 640 and 2,000 units.
6. Flow-design tools.
7. Customer contact and the service-system design matrix.
8. Customer-introduced variability and four strategies.
9. Poka-yokes (Three Ts), Ritz-Carlton, three service designs.
10. BPMN 2.0 essentials.
11. Service blueprint layers and lines.
12. KAI: the slowest stage sets capacity; baseline first.

Case studies

Case & worked answers

KAI — One Second at the Gambir Gate (with your syndicate's work)

Q1: the failure sits frontstage at the registration desk (causes backstage/support); it is felt at the line of interaction and gates don't touch it. Q2: gates 14,400/h, manual 540/h, registration 80/h; with 15% unregistered the ceiling is ~533/h, with 5% 1,600/h — registration binds. Q3: the missing exclusive gateway routes passengers before registration, with manual fallback. Q4: biometric = internet & on-site technology; manual = face-to-face tight specs; different skills; the manual path needs an SLA. Q5: your syndicate chose Option B (8 stations first) with three measures and pre-agreed thresholds — baseline before investing.

Worked examples

Problem 1 · Medium **Break-even pemilihan proses**

Proses A: biaya tetap Rp50.000, variabel Rp30/unit. Proses B: tetap Rp120.000, variabel Rp16/unit. Di volume berapa keduanya sama mahal? Mana yang dipilih untuk 8.000 unit?

1. Samakan: 50.000 + 30Q = 120.000 + 16Q.
2. 14Q = 70.000 → Q = 5.000 unit.
3. Di 8.000 unit: A = 50.000 + 240.000 = 290.000; B = 120.000 + 128.000 = 248.000.

Answer: Titik impas 5.000 unit. Untuk 8.000 unit pilih B (lebih murah Rp42.000).

Problem 2 Easy **Kapasitas sumber daya**

3 gate tiket, masing-masing 6 detik per penumpang. 2 meja registrasi, masing-masing 60 detik per penumpang. Berapa kapasitas per jam tiap sumber daya, dan mana bottleneck jika semua penumpang harus registrasi?

1. `Kapasitas = jumlah unit × 3.600 ÷ detik per penumpang`
2. Gate: 3 × 3.600 ÷ 6 = 1.800 orang/jam.
3. Registrasi: 2 × 3.600 ÷ 60 = 120 orang/jam.

Answer: Bottleneck = meja registrasi (120/jam). Sistem hanya bisa melayani 120 penumpang/jam walau gate sanggup 1.800.

Practice questions

Try answering first. The answer key is below each question.

1. Dell builds 384 configurations from 17 components. Environment…

A. MTS
B. ATO
C. MTO
D. ETO

Answer: B. Forecast modules, assemble to order.

2. At 1,200 units the cheapest option is…

A. Buy
B. Semi-automatic lathe
C. Machining centre
D. All equal

Answer: B. Between 640 and 2,000.

3. The line of visibility separates…

A. Customer and frontstage
B. Frontstage and backstage
C. Backstage and support
D. Evidence and journey

Answer: B. Below it can run like a factory.

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4. The BPMN gateway choosing exactly one branch is…

A. Parallel
B. Exclusive
C. Inclusive
D. Event-based

Answer: B. Exclusive.

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5. 15% must register, desks handle 80/h. Ceiling…

A. 80
B. 533
C. 1,600
D. 14,400

Answer: B. 80/0.15.

Sources

- P3MD MK4 OSCM — Meeting 6: Process Design and Analysis (class slides)
- Jacobs & Chase — OSCM Ch. 7 & 9; OMG BPMN 2.0; NN/g Service Blueprint 101; KAI case